



PRESTO GEOSYSTEMS

GENUINE GEOWEB® CELLULAR CONFINEMENT

Correlation of Subgrade Soil Strength Parameters for Cohesive Soils (Fine-Grained)

California Bearing Ratio	Undrained Shear Strength*	Hand Penetrometer Readings	Standard Penetration Resistance	Field Identification / Visual
<i>CBR (%)</i>	<i>C_u (kPa)</i>	<i>P_q (kg/cm²)</i>	<i>SPT (blows/300 mm)</i>	
< 0.4	< 11.7	< 0.25	< 2	Very Soft (extruded between fingers when squeezed), Man standing sinks >75 mm
0.4 – 0.8	11.7 – 24.2	0.25 – 0.50	2 – 4	Soft (molded by light finger pressure) Man walking sinks 50 -75 mm
0.8 – 1.6	24.2 – 47.6	0.50 – 1.0	4 – 8	Medium (molded by strong finger pressure) Man walking sinks 25 mm
1.6 – 3.2	47.6 – 95.9	1.0 – 2.0	8 – 15	Stiff (readily indented by thumb but not penetrated with great effort) Pick-up ruts 13 – 25 mm
3.2 – 6.4	95.9 – 191	2.0 – 4.0	15 – 30	Very Stiff (readily indented by thumb) Loaded dump truck ruts 25 – 75 mm
> 6.4	> 191	> 4.0	> 30	Hard (indented with difficulty by thumbnail) Loaded dump truck no ruts